

Anti-Money Laundering with Feedforward Neural Networks

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Abstract

This project explores the effectiveness of using a Feedforward Neural Network (FNN) to detect money laundering transactions, comparing its performance to widely used Graph Neural Network (GNN) architectures such as Graph Isomorphism Network (GIN) and Principal Neighbourhood Aggregation (PNA). While GNNs are often considered the standard for this task, they demand significant computational resources. In contrast, the FNN demonstrates competitive results, offering a lightweight and accessible alternative for institutions with limited computational capacity.

1. Introduction

1.1. Problem Domain

Money laundering is the process of concealing the origins of illegally obtained funds. The United Nations estimates that up to \$2 trillion USD is laundered each year. These funds often finance terrorism, drug trafficking, human trafficking, and the proliferation of weapons of mass destruction. Financial institutions are therefore under increasing pressure to identify and prevent illicit activities (UN 2023). Artificial Intelligence (AI) models capable of detecting such transactions can play a vital role in strengthening global financial security.

1.2. Background research

Due to privacy regulations, most banks cannot share real-world customer transaction data. This limits research in the field, with much of the current progress relying on synthetic datasets. One notable example is IBM's synthetic Anti-Money Laundering (AML) dataset (Erik Altman et al. 2024), which simulates real-world financial transactions and includes a small proportion of illicit activities. For this project, the HI-Small Trans.csv dataset from IBM was selected (E. Altman 2019), as it offers a balance of dataset size and proportion of laundering cases.

1.3. GNNs and FNNs in AML

Current consensus suggests that Graph Neural Networks (GNNs) are well-suited for AML detection, since financial

transactions can be represented as graph structures. However, GNNs require heavy computational resources, long training times, and additional preprocessing to convert data into graph form. This project challenges that assumption by applying a Feedforward Neural Network (FNN) to the same task, evaluating whether its simplicity and efficiency can still yield competitive results.

1.4. Overall results

The FNN achieved high accuracy in classifying legitimate transactions but struggled with illicit ones due to the highly imbalanced dataset. Nevertheless, it demonstrated clear advantages in terms of training speed and computational efficiency. For smaller institutions with limited infrastructure, the FNN offers a practical AML solution that can rival or even outperform certain GNN architectures.

2. Methodology

2.1. The algorithm

The Feedforward Neural Network (FNN) is a type of supervised artificial neural network where data flows in one direction, from the input layer through the hidden layers to the output layer. Each hidden layer consists of neurons that apply mathematical functions to detect patterns, with connections weighted to represent the strength of influence between neurons. Activation functions, such as sigmoid and ReLU, introduce non-linearity, allowing the network to capture complex relationships in the data (Dubey, Singh, and Chaudhuri 2021).

Two common activation functions were tested:

- **Sigmoid:**

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

- **ReLU:**

$$\text{ReLU}(x) = \max(0, x) = \frac{x + |x|}{2} = \begin{cases} x, & x > 0, \\ 0, & x \leq 0 \end{cases} \quad (2)$$

Training the network involves adjusting weights to minimise

prediction errors. For binary classification in this project, the binary cross-entropy loss is used:

$$L = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right] \quad (3)$$

The learning process can be summarised as follows. First, the final output of the network is calculated for each input, propagating signals through all hidden layers:

$$a^{(l)} = f(W^{(l-1)} \cdot a^{(l-1)} + b^{(l-1)}) \quad (4)$$

Next, the loss between the predicted and actual values is computed. Gradients of the loss function with respect to each weight are obtained through **backpropagation**, applying the chain rule to propagate errors from the output layer backwards through the network (Rumelhart et al. 1986). Finally, weights are updated using **gradient descent**, moving in the opposite direction of the gradient to reduce the loss:

$$W^{(l)} = W^{(l)} - \eta \frac{\partial L}{\partial W^{(l)}} \quad (5)$$

This process continues iteratively until the model converges to a state where it performs well on unseen data.

Although FNNs are not typically the first choice for money laundering detection, especially when the data is structured for Graph Neural Networks (GNNs), they provide notable advantages. FNNs require significantly less computational power and train much faster than GNNs, making them suitable for environments with limited resources. The time complexity of the final FNN model can be approximated as $O(N)$, where N is the number of neurons. In contrast, GNNs typically scale as $O(N^2)$, where N represents the number of nodes or vertices. This computational efficiency makes FNNs a practical alternative while still achieving competitive performance.

2.2. Dataset Preprocessing

The analysis begins with loading the HI-Small Trans.csv dataset, which was preprocessed using IBM's provided code. To reduce training time, a 25% sample of the full dataset was used. The categorical columns — including Sent Currency, Received Currency, and Payment Format — were transformed into a numerical format via one-hot encoding, allowing the neural network to process them effectively. Once the data was prepared, it was split into training, validation, and test sets in a 60–20–20 ratio. Finally, the features were standardised using StandardScaler, ensuring that each input variable contributed proportionally to the model's learning process.

2.3. Experimental Setup

The Feedforward Neural Network was trained over 10 to 20 epochs with a batch size of 32. During experimentation, several parameters were adjusted to optimise performance, including the number of epochs, hidden layer sizes, dropout rates, and activation functions. This process allowed the model to learn efficiently from the available data while reducing overfitting, particularly given the inherent class imbalance between regular and illicit transactions.

3. Evaluation Metrics

To assess the model's performance, multiple metrics were considered. Accuracy measured the proportion of correctly classified transactions, while the F1-score provided a balance between precision and recall, particularly important for the minority class of illicit transactions. The ROC AUC score offered an overall evaluation of the classifier's discriminative ability, and the confusion matrix gave detailed insight into where the model succeeded or struggled. These metrics were complemented by visualisations, including the ROC curve and a plotted confusion matrix, which helped in interpreting the model's practical effectiveness.

4. Results and Discussion

The baseline FNN, configured with three layers and trained for 10 epochs, achieved a remarkable accuracy of 99.91%, though the F1-score was 0.41, reflecting the challenge of detecting the rare illicit transactions. After tuning hyperparameters and increasing epochs, the optimised FNN reached 99.92% accuracy and improved the F1-score to 0.52. Examination of the confusion matrix confirmed that while legitimate transactions were almost perfectly identified, illicit transactions remained more difficult to classify due to the extreme class imbalance. Comparatively, simpler GNN models such as GIN and PNA achieved F1-scores of 0.29 and 0.57, respectively. This demonstrates that the FNN can match or outperform certain GNN approaches while benefiting from substantially reduced computational demands.

5. Conclusion

This project demonstrates that Feedforward Neural Networks provide a lightweight and practical approach for detecting money laundering. Despite the challenges posed by imbalanced datasets, the FNN performed competitively and offers a feasible alternative for institutions that cannot deploy computationally intensive GNNs. Future work could involve extending the FNN into a deeper architecture to capture more complex transaction patterns or integrating it with other models, such as LightGBM, to further improve the detection of illicit activities.

References

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